



Abhishek Janagoudar

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Master's student in Electrical and Information Technology, specializing in Automation and Robotics at Hochschule Darmstadt. I have three years of experience working with AVEVA Engineering and Electrical Administration. Actively looking for a working student or mandatory internship position, with the goal of continuing into a thesis opportunity.

🎓 EDUCATION

09/2024 – present	Master of Science (M.Sc) Electrical and Information Technology Darmstadt University of Applied Sciences Course Overview: Industrial Robotics, Safety In Industrial Automation, Computer Vision, Industry 4.0 Industrial Internet of Things (IIoT) Adaptive Control, Modelling and Identification, Advanced Programming Techniques, State Space Analysis, Human Machine Interface, Advanced Sensors	Darmstadt, Germany
07/2017 – 09/2021	Bachelor of Engineering (B.E) Electrical and Electronics The National Institute of Engineering Grade: 8.12 CGPA (German Grade: 1.7)	Mysore, India

📁 PROJECTS

Safety in Industrial Automation Lab: Programming, Evaluation, and Compliance

- Programmed Siemens S7-1516F safety PLC systems and ET200SP modules using TIA Portal, implementing critical safety logic and functions such as ESTOP1, TWO_H_EN, SFDOOR, and FDBACK.
- Conducted rigorous discrepancy analyses and black box testing to ensure safety integrity.
- Evaluated safety functions according to EN ISO 13849-1 standards using SISTEMA software, calculating Performance Levels (PLr), verifying compliance, and enhancing system reliability and safety.

Industry 4.0 & IIoT Lab: OPC UA, Node-RED & Virtual Commissioning

- Configured OPC UA servers on Siemens S7-1516F PLC, integrating OPC UA clients across various devices, including PC, tablets, and Node-RED IoT hubs.
- Developed interactive dashboards in Node-RED for real-time monitoring and control, emphasizing Industry 4.0 communication protocols.
- Conducted virtual commissioning with Siemens TIA Portal and PLCSim Advanced, validating automation logic through comprehensive digital twin simulations of sorting plant models.

Advanced Programming Technique Lab Projects

- BattleShip Game: Developed a console-based Battleship game in C++ to enhance programming skills, focusing on game logic and user interaction.
- Student Database: Developed and designed an optimized database system for university enrollment using C++ (object-oriented programming), ensuring efficient data management and retrieval.
- Reversi Board Game: Implemented a console-based Reversi board game in C++ to strengthen algorithmic thinking and data handling.

Robotics and Computer Vision Projects

- Designed and simulated a 3-DOF robotic arm with real-time motion control using Webots and integrated forward/inverse kinematics with DH parameters.
- Developed a LiDAR-based point cloud segmentation pipeline for indoor mapping using Python and ROS.
- Implemented visual servoing for Igus ReBeL 6DOF using OpenCV for template-matching and image-to-joint command mapping.
- Engineered forward and inverse kinematics solutions for UR5/UR10 robotic manipulators using MATLAB/Simulink.

SKILLS

Automation: TIA Portal, Siemens S7 Safety, PLC Programming, SISTEMA

Industrial Internet of Things (IIoT): OPC UA, Node-RED, MQTT, IoT Dashboards, IoT Gateway Integration, Digital Twins, Virtual Commissioning

CI/CD: (Continuous Integration/Continuous Deployment)

Robotics: Forward and Inverse Kinematics, Denavit-Hartenberg Parameterization, Robot Arm Kinematic Modeling, Simulation, Webots

Computer Vision: Image Processing, LiDAR, Point Cloud Processing.

Programming: Python, C++, SQL, Open CV

Control System: Matlab, Simulink

Soft Skills: Project Management, Agile/Scrum, Team Collaboration, Technical Training, Troubleshooting, Teamwork

Other: Safety in the Utility Industry, Operations, Maintenance, and Commissioning

Engineering Tools: AVEVA Engineering v15.x, AVEVA Electrical v12, Jira, Rally, Software testing

LANGUAGES

English

IELTS - 7.5 (C1)

Deutsch

A2.2

Hindi

PROFESSIONAL EXPERIENCE

02/2024 – 08/2024	J. Ray McDermott Engineering Services Private Ltd Associate System Engineer - AVEVA Electrical Administrator V12 - Administered AVEVA Electrical V12 for Ruya EPCIC offshore projects 11 & 13, delivering 50+ electrical datasheets and drawings on time. - Standardized project workflows that reduced rework hours and improved inter-team coordination. - Led tool training sessions, reducing internal troubleshooting incidents by 30% within 5 months.	Chennai, India
09/2021 – 02/2024	Infosys Ltd. Senior System Engineer - AVEVA Engineering V15 Electrical & Instrumentation Administrator - Led the administration and optimization of AVEVA Engineering 15.x for Electrical and Instrumentation projects. - Acted as Scrum facilitator, promoting Agile practices and coordinating sprint reviews, retrospectives, and daily stand-ups. - Championed Agile practices, integrating CI/CD tools - Managed overall application architecture, operations, and system maintenance for engineering tools. System Engineer: - Configured libraries and settings, including TAS, I/O cards, wiring rules, instruments, and cable catalogs. 50+ User Story deliverables (Electrical Load list, Cable Schedule, Reports, Datasheets, and Drawings). - Developed data model with classes, attributes, associations, LOVs database views, and Kraftwerk Kennzeichen System naming conventions. - Mentored 8 new engineers via recurring technical sessions on AVEVA Suite. - Performed various architectural activities like project handling, Builds, deployment, and set-up applications.	Mysore, India
07/2019 – 07/2019	Arkay Energy Ltd. Electrical Operations and Maintenance Intern Supported commissioning/testing of substation systems, performed O&M tasks, learned industrial safety.	Ramanathapuram, India
06/2019 – 06/2019	LARSEN AND TOUBRO ELECTRICALS Electrical Commissioning Intern Assisted in electrical system setup/testing, troubleshooting, and system optimization.	Khargone, India